Bridges Across Cuisines

Mapping the residual geography
of global ingredient similarity

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Introduction

Thai and Vietnamese cuisines, separated by roughly 700 km of mainland Southeast Asia, share an ingredient profile resemblance well above the global mean. British and Russian cuisines, separated by 2,500 km and divided by language, climate, and political history, share a resemblance higher than most pairs of European neighbors. Filipino and Spanish cuisines, separated by more than 13,000 km across the Pacific and the Atlantic, share a resemblance comparable to that of regional neighbors elsewhere on the map.

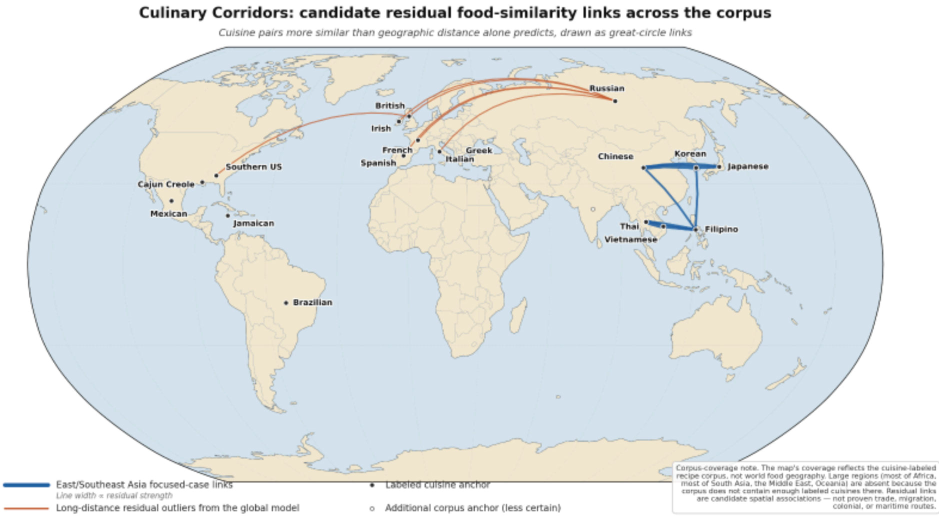
These three pairs are not curiosities. They are systematic. Across the world's documented cuisines, a striking fraction of the variation in ingredient resemblance cannot be explained by geographic distance. Some pairs are far apart yet remarkably similar. Some pairs sit inside the same region yet diverge sharply. The geography that organizes cuisine is not the geography of straight-line kilometers.

Most analyses of food treat cuisine as a cultural object: tastes, traditions, identities. Most spatial analyses, when they reach into food at all, treat distance as the primary explanatory variable for resemblance. This project sits at the intersection. It measures pairwise ingredient similarity systematically, models how similarity declines with geographic distance, and then asks where that decline does not hold. The leftover — residual cuisine similarity — turns out to be where the geography of food becomes interesting.

The central question is:

> *Once geographic distance is accounted for, where does cuisine resemblance still exceed what proximity predicts – and what does that residual structure reveal about how cuisines actually connect?*

The map below is the project's starting point. Cuisine pairs whose ingredient similarity exceeds distance-based expectation are drawn as great-circle links between approximate cuisine anchors on a Robinson-projection world map. Some links connect adjacent regional neighbors — the kinds of resemblance distance does explain. Others span continents and oceans, connecting cuisines that proximity alone would never predict to share. The latter are where this project's analytical interest lies.



Candidate residual cuisine corridors across the project corpus. Blue great-circle links mark the East/Southeast Asia focused case (line width proportional to residual strength); orange links mark long-distance residual outliers — pairs sitting well above the distance-similarity regression line. The map's geographic coverage reflects the cuisine-labeled recipe corpus, not world food geography. Residual links are candidate spatial associations, not proven exchange routes.

Robinson-projection world map with beige land and light-blue ocean basemap, country borders, and a subtle 30-degree graticule. Blue great-circle lines connect East and Southeast Asian cuisine anchors at varying widths. Orange great-circle lines connect long-distance pairs including British–Russian, Irish–Russian, French–Russian, Italian–Russian, and British–Southern US. Cuisine anchors in Europe, North America, the Caribbean, and South America are marked and labeled. A boxed corpus-coverage note in the lower right names the regions absent from the corpus.

The sections that follow describe how the analysis was built and what the residual network looks like once distance is removed. The findings work outward from the underlying baseline to two complementary structural views — a spatial-grouping view that reveals where residuals concentrate by configuration of cuisine pairs, and a network-position view that identifies which individual cuisines anchor the residual structure most actively. The strongest focused regional case sits in East and Southeast Asia, where mainland adjacency, peninsular geography, and archipelagic structure combine to produce the most cleanly readable corridor in the corpus. Four cuisine cases — two Asian, two non-Asian — close the analysis, each illustrating a distinct structural role.

How the analysis works

The analysis proceeds in three stages, each producing a layer that the next builds on.
Stage 1: Build comparable cuisine profiles. Cuisine-labeled recipes from a large recipe corpus [1, 2] are normalized so closely related ingredient names map to a single canonical ingredient — "scallion" and "green onion" become one entry. Ingredients are then aggregated into a cuisine-by-ingredient matrix in which each cuisine is represented as a frequency vector across thousands of normalized ingredients. This is the project's textual, chemical, and culinary signature for each cuisine.
Stage 2: Measure ingredient similarity and geographic distance. Pairwise cuisine similarity is computed using cosine similarity on the ingredient frequency vectors [3], with generic-ingredient filtering and robustness checks. Each

cuisine label is assigned an approximate geographic anchor — a centroid representing the cuisine's home territory. Pairwise geodesic distances between anchors are computed using GeoPy's great-circle method [8].
Stage 3: Extract the residual. Cuisine similarity is regressed on log geographic distance. The fitted line gives, for each pair, a predicted similarity based on distance alone. The residual — observed similarity minus distance-predicted similarity — becomes the analytical object. Positive residuals identify pairs more similar than distance predicts. Mapped, they form a candidate network of long-distance culinary connection.
The four findings that follow draw on this residual layer in different ways: a distributional view, a spatial-grouping view, a network-position view, and a focused regional case.

> *Residuals are the hinge of the project. They turn ingredient resemblance into a spatial question.*

Finding 1: Distance shapes cuisine similarity, but explains less than half of it.

Ingredient similarity declines with geographic distance, but the decline is shallow. The fitted relationship across all pairwise cuisine comparisons returns a slope of −0.116 per unit of log distance and an R^2 of 0.355. In concrete terms: doubling the distance between two cuisines reduces their predicted ingredient similarity by approximately 0.080 cosine-similarity units — meaningful, but small relative to the range of observed similarities, which spans roughly 0.05 to 0.92. Roughly 64 percent of the variation in cuisine similarity is structured but not explained by distance.
The shallowness matters analytically. If distance explained 90 percent of cuisine resemblance, a residual analysis would reveal little: the line would absorb most of the signal and leave only noise behind. At $R^2 = 0.36$, the residuals carry the majority of the spatial information. The largest positive residuals — pairs where similarity exceeds distance-based

expectation by 0.2 cosine units or more — include several long-distance combinations the distance baseline would never anticipate: British–Southern U.S., British–Russian, Irish–Russian, French–Russian, Italian–Russian. These pairs sit between 2,500 and 8,000 km apart yet retain ingredient similarities that match or exceed those of regional neighbors. The figure below shows the underlying pattern. Each point is one cuisine pair; the regression line is the distance baseline; named labels mark the strongest positive residuals discussed above.

Distance baseline for cuisine similarity. Each point is a pairwise cuisine comparison: cosine similarity of filtered ingredient profiles plotted against log geographic distance between cuisine anchors. The fitted line — similarity = 1.273 − 0.116 × log(distance), $R^2 = 0.355$ — defines the distance-only expectation. Pairs above the line are positive residuals; the labeled points (Thai–Vietnamese, British–Russian, Irish–Russian, French–Russian, Italian–Russian, British–Southern U.S.) are the strongest positive residuals in the corpus.

Scatter plot of cosine similarity of filtered cuisine ingredient profiles versus log geographic distance between cuisine anchors. A regression line of similarity = 1.273 − 0.116 × log(distance) is shown with $R^2 = 0.355$. Points above the line, including Thai–Vietnamese at the short-distance end and British–Russian, Irish–Russian, French–Russian, Italian–Russian, and British–Southern U.S. at the long-distance end, are highlighted as positive residuals.

> *Two-thirds of cuisine similarity sits outside what distance alone can explain. That two-thirds is where the analysis goes next.*

Finding 2: Residual cuisine similarity is structured by spatial grouping – and the ranking inverts the obvious intuition.

Finding 1 establishes that residuals exist and are sometimes very large. Finding 2 asks whether those residuals concentrate in any particular configuration of cuisine pairs.

To test this, pairs are partitioned into five spatial groupings based on the regional relationship between the two cuisine anchors: same-subregion (the tightest geographic neighbors), same-region cross-subregion (still within the same broad continental region but a different subregion), East/Southeast Asia cross-subregion (the focused regional case examined in Finding 4), Iberian/Atlantic interregional (the long-distance Iberian–Atlantic–Pacific cluster of cuisines), and other cross-region (everything else). Mean residual cuisine similarity is then computed within each grouping.

A distance-only intuition would predict a monotonic decline: same-subregion pairs at the top, intermediate groupings in the middle, distant cross-region pairs at the bottom. The data refuse to follow that order.

Mean residual cuisine similarity by spatial grouping. The Iberian/Atlantic interregional grouping sits at the top of the ranking (+0.139, n = 11), exceeding the same-subregion baseline (+0.115, n = 11). All other groupings sit slightly below the distance-only expectation. The headline annotation flags the project's clearest non-regional finding: long-distance Iberian–Atlantic–Pacific pairs are, on average, more similar than distance alone predicts by a wider margin than even the closest geographic neighbors.

Horizontal bar chart titled "Where does residual cuisine similarity concentrate? Mean residual by spatial grouping." Five bars from top to bottom: Iberian/Atlantic interregional at +0.139 (n=11) in saturated orange, Same subregion at +0.115 (n=11) in lighter orange, Same region cross-subregion at -0.011 (n=32), East/SE Asia cross-subregion at -0.014 (n=9), Other cross-region at -0.020 (n=127). A vertical zero line is labeled "distance-only expectation." A boxed callout to the right reads "Highest mean residual in the prototype. This is the project's mandatory non-Asia diagnostic case."

The Iberian/Atlantic interregional grouping has the highest mean residual in the corpus at +0.139 (n = 11). The same-subregion grouping comes second at +0.115 (n = 11). Same-region cross-subregion (n = 32), the East/Southeast Asia cross-subregion (n = 9), and other cross-region pairs (n = 127) sit slightly below the distance-only baseline at -0.011, -0.014, and -0.020 respectively. The full top-to-bottom ranking is, therefore, not the geography of adjacency. It is the geography of a specific long-distance configuration that connects Iberian, Atlantic-rim, and Pacific-archipelagic cuisine anchors.
The cuisines participating in the Iberian/Atlantic interregional

grouping include Spanish, Filipino, Mexican, Cajun/Creole, Brazilian, Jamaican, and Southern U.S. The pairs involving these labels span three continents and two oceans — and yet, on average, exceed the distance-only baseline by a wider margin than the typical same-subregion pair. The magnitude of that difference is concrete: the Iberian/Atlantic mean exceeds the "other cross-region" mean by 0.159 cosine units, which is approximately twice the similarity loss that a doubling of geographic distance would predict from the Finding 1 baseline.

This is the project's first analytically distinctive finding. It is not that distance fails to predict similarity in general — it does, weakly. It is that the specific long-distance pairs connecting Iberian, Atlantic, and Pacific cuisines are systematically more similar than distance permits. Something else is structuring this configuration.

> *The strongest residuals in the corpus are not concentrated among regional neighbors. They are concentrated in a specific Iberian–Atlantic–Pacific configuration that distance alone would not predict.*

Finding 3: A small set of bridge cuisines anchors the residual network — and they span Asian, Eurasian, and Atlantic geographies in distinct structural roles.

Findings 1 and 2 work at the pair level. Finding 3 aggregates from pairs to cuisines.

For each cuisine, a residual bridge score is computed by combining five components extracted from the residual matrix: positive residual degree (how many other cuisines the focal cuisine connects to with a positive residual), participation in the corpus's top-strength residual links, mean residual magnitude across the cuisine's pairs, long-distance residual score (whether the cuisine's residual partners are geographically distant or near), and overall residual behavior. Each component is normalized to a 0–1 scale and the five are combined with equal weights. The result is a single per-cuisine score capturing how actively each cuisine functions as a connector in the residual network.

The top ten bridge cuisines, ranked by score:

1. Filipino — 0.872.
2. Russian — 0.843.
3. Southern U.S. — 0.694.
4. Jamaican — 0.685.
5. French — 0.656.
6. Spanish — 0.537.
7. British — 0.518.
8. Irish — 0.449.
9. Italian — 0.3210.
10. Brazilian — 0.31

The ranking has structure worth reading carefully. Filipino sits at the top, anchoring a Pacific-archipelagic role that connects East and Southeast Asian cuisines through maritime geography while also extending across the Pacific to Spanish-speaking partners. Russian sits second, anchoring a continental Eurasian span whose strongest residual partners are European cuisines on the western edge of that span. The remaining eight cuisines cluster geographically along the Atlantic basin and its connected geographies — the British and Irish Isles, the Iberian and French peninsulas, the Italian peninsula, the Caribbean, the Southern U.S. coast, and Brazil's Atlantic shore. The bridge network, in other words, has three structural geographies: a Pacific-archipelagic node, a Eurasian continental span, and a dense Atlantic-rim cluster.

Two-panel residual bridge index. Left: Robinson world map of cuisine anchors with circle size proportional to bridge score, showing the Pacific-archipelagic node (Filipino), the Eurasian continental anchor (Russian), and the Atlantic-rim cluster (British, Irish, French, Spanish, Italian, Southern U.S., Jamaican, Brazilian). Right: ranked bar chart of the top ten bridge scores. The figure makes the network's three structural geographies visible at a glance.

Side-by-side panels. Left panel is a Robinson-projection world map with beige land and light-blue ocean. A large blue circle marks Filipino in the western Pacific; large orange circles mark Russian in central Asia, Southern US in the U.S. southeast, Jamaican in the Caribbean, French, Spanish, British, Irish, and Italian in western and southern Europe, and Brazilian in central South America. Right panel is a horizontal bar chart titled "Top 10 bridge cuisines" showing residual bridge scores from 0.87 (Filipino, blue) to 0.31 (Brazilian), with all other bars in orange.

The bridge network is also concentrated. The top three cuisines — Filipino, Russian, and Southern U.S. — together account for 41 percent of the total bridge weight in the top 10. The top five — adding Jamaican and French — account for 64 percent. The residual network is not flat; it is anchored by a small, geographically specific set of cuisines that connect the rest into a coherent long-distance structure.

It is worth being precise about what a bridge score does and does not say. It is a network position. It says that, after the distance baseline is removed, the focal cuisine participates in many strong residual links with cuisines that are not its geographic neighbors. It does not assert that the focal cuisine caused those resemblances or that any one cuisine "owns" a corridor. It reports a structural fact about the corpus: ingredient resemblance, when distance is controlled for, organizes itself around a small number of high-connectivity nodes whose geographies are distinct rather than redundant. The three structural geographies — Pacific-archipelagic, Eurasian continental, and Atlantic-rim — each play a different role. The Filipino node connects East/Southeast Asian regional cuisines (the focused case in Finding 4) into a wider global network. The Russian node connects continental Eurasia to the western European Atlantic cluster. The Atlantic-rim cluster contains the densest concentration of mutually high-residual pairs, which is consistent with Finding 2's identification of the Iberian/Atlantic interregional grouping as the highest-mean-residual configuration in the corpus. The bridge structure and the spatial-grouping structure tell consistent stories from different angles.

> *The residual network has three distinct geographies – Pacific-archipelagic, Eurasian continental, and Atlantic-rim – each anchored by different cuisines and each playing a different structural role.*

Finding 4: The strongest focused residual corridor sits in East and Southeast Asia – and combines mainland, peninsular, and archipelagic structure.

Finding 3 identifies which cuisines connect the residual network most actively. Finding 4 asks where the residual links concentrate spatially when narrowed to a single focused regional case. Among focused cases, East and Southeast Asia produce the strongest, cleanest residual corridor in the corpus.

The strongest single residual link in the entire dataset is Thai–Vietnamese at +0.359 — a mainland adjacency case where ingredient similarity exceeds distance-based expectation by the largest margin observed in the corpus. Chinese–Korean follows at +0.306, a regional proximity case across the Yellow Sea. Filipino–Thai (+0.219) and Filipino–Vietnamese (+0.209) span the South China Sea and represent island-maritime links that distance alone would predict to be far weaker. Korean–Japanese (+0.20) closes the top five.

The East/Southeast Asia focused-case map shows the strongest regional residual cuisine links over real geography — country fills, coastlines, rivers, and the South China Sea / Sea of Japan. Link color encodes spatial type: dark blue for mainland adjacency, teal for regional proximity, magenta for island/maritime. Line width is proportional to residual strength; the strongest residual values are labeled inline. A side panel ranks the top five focused-case residual links.

Regional map of East and Southeast Asia rendered in PlateCarree projection from 93°E to 145°E and 4°N to 50°N, with beige land, light-blue ocean, faint country borders, rivers, and a lat/long graticule. Six cuisine anchors are marked: Chinese, Japanese, Korean, Thai, Vietnamese, and Filipino. Colored great-circle links connect them: dark-blue Thai–Vietnamese labeled r=0.36 (mainland adjacency); teal Chinese–Korean labeled r=0.31 and Korean–Japanese labeled r=0.20 (regional proximity); magenta Filipino–Thai labeled r=0.22 and Filipino–Vietnamese labeled r=0.21 (island/maritime). A right-side panel ranks the top 5 residual links with their residual values.

Three structural features make the East/Southeast Asia case analytically clean. First, the top residuals span three distinct link types: mainland adjacency (Thai–Vietnamese), regional proximity (Chinese–Korean, Korean–Japanese), and island-maritime (Filipino–Thai, Filipino–Vietnamese). The corridor is not a chain of neighbors — it is a small network with multiple geometric configurations represented. Second, residual strength is high: all five top links sit above +0.20, well above the corpus-wide same-subregion mean of +0.115. Third, the underlying geography is interpretable: mainland Indochina, the East China Sea, the Korean peninsula, the Japanese archipelago, and the Philippine archipelago appear on the same map and the residuals can be read against their physical structure.

Adding topographic and maritime context makes the corridor's structure more legible without changing any of the residual values. The map below shows the same focused case overlaid on shaded relief and coastlines.

The Run 5 relief map places the strongest East/Southeast Asia residual links over topographic, coastal, island, and maritime context. Line width reflects residual strength; line color indicates same-subregion, island/maritime, or cross-subregion link types. The map clarifies the spatial reading of the corridor — peninsulas, archipelagos, shallow seas, and mountain barriers — without claiming that any of these features causally produced the observed resemblance.

Shaded relief and coastline map of East and Southeast Asia. Cuisine anchors for Chinese, Korean, Japanese, Thai, Vietnamese, and Filipino are connected by colored corridor lines. Annotations call out the Tibetan Plateau and Himalayan barrier to the west, peninsula and island exchange context for Korea and Japan, the South China Sea maritime context for Filipino-Vietnamese-Thai links, and mainland Southeast Asia adjacency context for Thai-Vietnamese. A side panel lists top residual links with numerical scores.

Three features become more visible on the relief map. The Tibetan Plateau and Himalayan barrier mark the western edge of the corridor — the regions immediately west of the East/SE Asia residual network are separated from it by some of the highest terrain on Earth, which corresponds with the absence of strong residual links across that boundary. The Korean peninsula, Japanese archipelago, and Philippine archipelago form a coastal-and-island ring that the East Asian residuals trace closely; the Filipino node, in particular, makes more sense as an archipelagic bridge than as a point on a continental map. The South China Sea and Sea of Japan act as connectors rather than separators — maritime space across which residual similarity is actively maintained.

This is consistent with what Finding 3 implies more globally. Bridge cuisines tend to sit at the intersection of land and water — at archipelagic nodes (Filipino), at peninsular extensions (Iberian), at island-and-coast complexes (British, Irish, Italian), or at long continental spans with coastal access at multiple ends (Russian). The strongest residual network in the corpus does not respect simple straight-line distance. It rides the physical structure of coastlines, peninsulas, and shallow seas.

> *The corridor is not a chain. It is a small network where mainland, peninsula, and archipelago each contribute a distinct kind of residual link.*

Four cuisines that explain the pattern

The four findings above describe the residual geography in aggregate. The four cuisines that follow illustrate how that geography concentrates around individual cuisine anchors. Each represents one structural role in the network. Two are Asian — Filipino and Thai — and they anchor the Pacific-archipelagic and East/Southeast Asian regional structures respectively. Two are non-Asian — Russian and Spanish — and they anchor the Eurasian continental and the Iberian/Atlantic–Pacific structures respectively.
Together, the four describe the shape of the residual network from four complementary vantage points: an archipelagic bridge, a continental bridge, a regional hub, and a long-distance Iberian–Atlantic–Pacific node.

Filipino: archetypal archipelagic bridge

Bridge score: 0.87 (rank 1 of 10). Location: Philippine archipelago. Top residual links: Thai (+0.219), Vietnamese (+0.209), Korean (+0.12), Chinese (+0.11), plus participation in the long-distance Iberian–Atlantic–Pacific group anchored at the Spanish end.
Filipino is the highest-scoring bridge cuisine in the entire corpus by a clear margin — its bridge score of 0.87 alone accounts for 15 percent of the total weight held by the top ten. The cuisine sits in the Philippine archipelago, geographically positioned between mainland Southeast Asia, East Asia, the Pacific Ocean, and — through historical exchange — the Spanish-speaking world.
The residual structure of Filipino cuisine reads as a maritime bridge. Its links to Thai and Vietnamese are island-to-mainland connections across the South China Sea. Its links to Chinese and Korean are island-to-mainland connections across the East and Yellow Seas. Its participation in the Iberian/Atlantic interregional grouping (Finding 2) extends the network across the Pacific. No other cuisine in the corpus participates in residual links across this many distinct geographic configurations, and no other cuisine combines a regional Asian role and a trans-Pacific role with comparable

strength.

The Filipino node is what makes the East/Southeast Asia case (Finding 4) and the Iberian/Atlantic–Pacific finding (Finding 2) connect. It is the cuisine where the corridor structure meets the bridge structure — the analytical hinge between the project's regional and global scales.

Russian: long-distance Eurasian bridge

Bridge score: 0.84 (rank 2). Location: continental Eurasia. Top residual partners: British, Irish, French, Italian, Spanish, Southern U.S. — every one of them named in the long-distance positive-residual cluster of the distance-similarity scatter (Figure in Finding 1).

Russian sits second only to Filipino on bridge score, and the comparison between the two is structurally informative. Where Filipino bridges across maritime space, Russian bridges across continental Eurasian distance. Its top residual partners are all European or Atlantic-linked cuisines at distances ranging from roughly 2,500 to 8,000 km. None of these pairs would be predicted to share strong ingredient resemblance based on distance alone — they sit well above the regression line in Finding 1 — yet they consistently do.

The Russian span is geographically distinctive in two respects. First, the country's territorial extent reaches from the Baltic to the Pacific, and historical Russian and Soviet trade and migration networks connected northern and continental Europe to Central Asia and the Pacific Far East. Second, the residual partners cluster at one end of that span — the European end — rather than spreading across the full Eurasian range. Russian's bridge structure is a long-distance European connector more than a Eurasian span as such.

This is the case that most clearly demonstrates that bridge scores are not regional reach in disguise. Russian's bridges are intercontinental, and its continental-bridge geometry is the structural mirror image of Filipino's maritime-bridge geometry. The two highest-scoring bridges in the corpus do the same job through completely different geographies.

Thai: regional hub at the heart of the strongest corridor

Top residual links: Vietnamese (+0.359 — strongest single link in the corpus), Filipino (+0.219), plus secondary links to neighboring E/SE Asian cuisines. Location: mainland Southeast Asia. Bridge score: not in the top 10.

Thai is not a top-ten bridge cuisine. Its analytical role is different: it sits at the center of the strongest focused residual corridor in the corpus. The Thai–Vietnamese link is the highest-residual pairwise link in the entire project at +0.359, a mainland adjacency case where ingredient similarity exceeds distance-based expectation by the largest margin observed.

The Thai-anchored sub-network is dense rather than far-reaching. Thai connects to Vietnamese (mainland adjacency, the strongest case), Filipino (island-maritime, the strongest island-mainland case), and other neighboring cuisines in the regional mesh. These are short-distance links — exactly the kind distance-based intuition expects to find positive residuals — but the magnitudes exceed what a typical same-subregion pair shows globally (mean +0.115 across the corpus).

If Filipino represents the bridge — high cross-region reach — Thai represents the hub: high regional concentration. The project's strongest focused-case residuals are produced by the interaction of both roles. The corridor is anchored by a regional hub and connected outward through a maritime bridge.

Spanish: Iberian/Atlantic–Pacific node

Bridge score: 0.53 (rank 6). Location: Iberian peninsula. Top residual partners: Filipino, Mexican, Brazilian, Cajun/Creole, Jamaican — the cuisine pairs that drive the Iberian/Atlantic interregional grouping (Finding 2). Russian also appears among Spanish's labeled long-distance positive-residual partners in the scatter view (Finding 1).

Spanish sits at one end of the project's most analytically distinctive cluster: the Iberian/Atlantic interregional grouping with the highest mean residual in the corpus (+0.139, n = 11). Spanish cuisine's residual links span the Atlantic and the Pacific. The distances are large — the link to Filipino covers roughly 13,000 km — and the ingredient resemblance is consistent.

Among the Iberian/Atlantic group, Spanish is the cuisine whose residual links extend farthest geographically. Its Pacific-spanning connection to Filipino, in particular, is a residual that would be invisible in a distance-only view of cuisine similarity but becomes the highest-magnitude long-distance configuration in the corpus once distance is removed. The Spanish node anchors the project's clearest evidence that long-distance residual cuisine similarity is not noise. It is a structured pattern with identifiable geographic configurations.

Together, Filipino, Russian, Thai, and Spanish span the residual geography of the corpus. Filipino is the maritime bridge. Russian is the continental bridge. Thai is the regional hub. Spanish is the long-distance Iberian/Atlantic–Pacific node. Each occupies a different role; together they describe the structure of the residual network the project set out to reveal.

> *Four cuisines, four roles. The residual network has a shape, and these four anchors are what give it that shape.*

Conclusion

The simplest measurement in this project — distance from one cuisine to another — explains roughly a third of the variation in pairwise ingredient similarity. The remaining two-thirds is where the project's analytical interest lives, and that two-thirds turns out to be structured rather than random. The structure has a specific shape. It concentrates around a small number of bridge cuisines whose geographies are distinct rather than redundant: a Pacific-archipelagic anchor (Filipino), a Eurasian continental anchor (Russian), and a dense Atlantic-rim cluster (the British and Irish Isles, the Iberian and French peninsulas, the Italian peninsula, the Caribbean, the Southern U.S. coast, and Brazilian Atlantic shore). It clusters into specific spatial configurations — most strikingly an Iberian/Atlantic interregional pattern that exceeds the same-subregion baseline. It produces a strongest focused regional corridor in East and Southeast Asia where mainland adjacency, peninsular geography, and archipelagic structure combine. And it suggests, by the diversity of these structural geographies, that the geography organizing cuisine resemblance is not the geography of straight-line distance. It is the geography of long-range exchange networks, maritime corridors, archipelagic bridges, peninsular extensions, and historical contact zones whose imprint on ingredient profiles persists long after the original conditions that produced them.

The strongest contribution of this project is to surface the structure that distance alone obscures. Cuisine, viewed through residuals, has a network shape — and the shape spans Asian, Eurasian, and Atlantic geographies in roles that complement rather than substitute for one another.

For food-systems researchers, agricultural economists, and historians of exchange, the residual network is a starting point for more specific inquiry: which historical events, ecological conditions, or trade networks correspond with which residual links? For methodologists, the bridge index demonstrates that aggregating pairwise residuals to the unit (cuisine) level surfaces structural roles that pairwise analysis alone cannot.

For curious general readers, the message is simpler. Cuisine resemblance is not just about who lives next door. It is about who has been connected, and the network of that connectedness has a shape that a map can show.

Cuisine, viewed through residuals, has a network shape that spans Asian, Eurasian, and Atlantic geographies in distinct structural roles — and the shape is invisible in any analysis that treats distance as the only spatial variable.

Sources

This project was developed in Python and ArcGIS Pro / ArcGIS Online, with figure rendering in Matplotlib and Cartopy.

Data processing. Recipe corpus normalization and the cuisine-by-ingredient matrix were built in Python using Pandas and NumPy. Cosine similarity was computed using SciPy. Geodesic distances between cuisine anchors were computed using GeoPy's great-circle distance method. Distance-residual modeling used a simple linear regression on log distance, implemented in scikit-learn.

Spatial analysis and visualization. ArcGIS Pro provided spatial joins, projection, and the regional case-study working layers. ArcGIS Online hosted the residual link layers used during analysis. Final cartographic figures — the global hero map, the East/Southeast Asia regional map, the bridge index, and the spatial-grouping bar chart — were rendered in Python using Matplotlib and Cartopy with Natural Earth basemap data at 110m and 50m resolution.

Bridge index. The residual bridge score for each cuisine combines five components computed from the residual matrix: positive residual degree, participation in the corpus's top-strength residual links, mean residual magnitude, long-distance residual score, and overall residual behavior. Each component is normalized to a 0–1 scale; the five are combined with equal weights to produce the published bridge score.

Boundary/permeability test. Cuisine pairs were partitioned into five spatial groupings based on the regional and subregional relationships between the two anchors. Mean residual within each grouping was computed and compared.

Data sources

The recipe corpus is the Yummly "What's Cooking" Kaggle dataset, accessed via the prepared version in David Zelený's anadat-r repository [1, 2]. The dataset contains 39,774 cuisine-labeled recipes, 20 cuisine labels, and 6,714 distinct raw ingredient names. Cuisine labels are treated as approximate cultural-geographic anchors rather than exact nation-state polygons; some labels (Southern U.S., Cajun/Creole) name regional or diasporic cuisines, while others (Chinese, Indian, Russian) name large national or civilizational categories. The corpus is platform-mediated and U.S.-recipe-skewed, which is one reason the project's strongest focused inference sits in cuisines for which the corpus has dense, internally consistent ingredient coverage. Generic pantry ingredients were filtered before similarity computation, and an alias crosswalk normalized closely related ingredient name variants to canonical forms.

Each cuisine label was assigned an approximate geographic anchor — a centroid representing the cuisine's home territory — with coordinates documented in the project repository. Pairwise geodesic distances between anchors were computed using GeoPy's great-circle method on the WGS84 ellipsoid [8]. Pairwise cuisine similarity uses cosine similarity on the cuisine-by-ingredient frequency matrix, with the data pipeline and similarity model implemented in Python using NumPy, pandas, and scikit-learn [4, 5, 6, 7].

Cartographic basemap data are drawn from Natural Earth (v5.1.1) at 110m resolution for global maps and 50m for regional maps, providing land, ocean, country borders, coastlines, rivers, and lakes layers [13]. Regional groupings used in the spatial-grouping analysis (Finding 2) follow the United Nations M49 standard for statistical regions and subregions [14]. The Run 5 topographic corridor map (Finding 4, second figure) draws on the ETOPO 2022 15-arc-second Global Relief Model from NOAA NCEI [15]. Final cartographic figures were rendered in Python with Matplotlib and Cartopy [9, 10]; earlier renderings used the Matplotlib Basemap toolkit [11], which has since been deprecated in favor of Cartopy. ArcGIS Pro and ArcGIS Online provided spatial joins, projection management, and the StoryMap publication platform [12].

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